

13 The ideal gas undergoes an expansion.

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[Turn over

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Work done on gas, $W = -p\Delta V \approx -29.9 \text{ J}$

By first law of thermodynamics,

$$\Delta U = Q + W$$

$$= 24 + (-29.9)$$

$$= -5.9 \text{ J}$$

Answer: B

1.4. With respect from origin